3) Using Master method the $T(n) = 2T\left(\frac{n}{2}\right) + n \log n$ then T(n) is

a. $f(n) = \Theta(n)$
b. $f(n) = \Theta(n \log n)$
c. $h(n) = \Theta(\log n)$
d. Can't solve

4) If you have a 0/1 Knapsack problem with 6 items, how many potential item combinations must be examined using the brute-force approach?

a. 6
b. 12
c. 32
d. 64

5) Which technique efficiently solves optimization problems by breaking them into smaller overlapping subproblems while avoiding redundant calculations?

a. Divide and conquer
b. Branch and bound
c. Dynamic programming
d. Greedy

6) In the context of the graph coloring problem, which of the following statements represents an optimization problem?

a. For a graph G=(V,E) and a given integer k, can G be colored using k colors?
b. Determine the chromatic number for a graph G=(V,E).

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5) **Which technique efficiently solves optimization problems by breaking them into smaller overlapping subproblems while avoiding redundant calculations?**
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   b. *Branch and bound*
   c. *Dynamic programming*
   d. *Greedy*
6) **In the context of the graph coloring problem, which of the following statements represents an optimization problem?**
   a. *For a graph G=(V,E) and a given integer k, can G be colored using k colors?*
   b. *Determine the chromatic number for a graph G=(V,E).*
   c. *For a graph G=(V,E) and a given integer k, how many distinct k-colorings are possible?*
   d. *None of the above.*
7) **Shortest path to all the vertices of a graph from a given vertex of the same graph is obtained by:**
   a. *Kruskal's algorithm*
   b. *BFS*

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b. Determine the chromatic number for a graph G=(V,E).
c. For a graph G=(V,E) and a given integer k, how many distinct k-colorings are possible?
d. None of the above.

7) Shortest path to all the vertices of a graph from a given vertex of the same graph is obtained by:
a. Kruskal's algorithm
b. BFS

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c. Dijkstra's algorithm
d. DFS

8) Breadth First Search graph algorithm uses ——— data structure for its implementation:
a. Queue
b. Stack

c. *Dijkstra's algorithm*
d. *DFS*

8) ***Breadth First Search graph algorithm uses ——— data structure for its implementation:***
a. *Queue*
b. *Stack*
c. *Tree*
d. *All the above answers are incorrect.*

9) ***Consider a list of 10 elements :***
list = [7,11,3,10,17,23,1,4,21,5]
Determine the partially sorted list after three compete passes of insertion sort in descending order
a. *11,10,7,3,17,23,1,4,21,5*
b. *11,17,23,10,7,21,5,4,3,1*
c. *3,7,11,10,17,23,1,4,21,5*
d. *23,21,17,10,3,7,1,4,11,5*
e. *None of the above.*

10) - ***What is the recurrence relation of Merge Sort?***
a. $T(n) = T(n-2) + O(n)$
b. $T(n) = T(n-1) + O(n)$
c. $T(n) = 2T(n/2) + O(n)$
d. $T(n) = T(n/2) + T(n) + O(1)$

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1) **What is the closet time complexity of the function $f(n) = n \log 4 + \log n$:**

a. $O(n)$
b. $O(n \log n)$
c. $O(\log n)$
d. $O(n \log (n^2))$

2) **To which class does the CNF-satisfiability problem belong?**

a. *NP.*
b. *P.*
c. *NP-complete.*
d. *NP-hard.*
e. *All answers are incorrect.*

3) *Using Master method the* $T(n) = 2T\left(\frac{n}{2}\right) + n \log n$ *then T(n) is*

a. $f(n) = \Theta(n)$
b. $f(n) = \Theta(n \log n)$
c. $h(n) = \Theta(\log n)$
d. *Can't solve*

4) ***If you have a 0/1 Knapsack problem with 6 items, how many potential item combinations must be examined using the brute-force approach?***

a. *6*
b. *12*
c. *32*

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6) In the context of the graph coloring problem, which of the following statements represents an optimization problem?

a. For a graph G=(V,E) and a given integer k, can G be colored using k colors?
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A. For each pair of functions f(n) in rows and g(n) in columns in the table below, write O, Ω, or Θ in the appropriate space, depending on whether f(n) = O(g(n)), f(n) = Ω(g(n)), or f(n) = Θ(g(n)). If there is more than one relation between f(n) and g(n), write only the strongest one. The first line is a demo solution. We use lg to denote the base-2 logarithm. [3 marks]

	n	$n \lg n$	n^2
$n \lg^2 n$	Ω	Ω	O
$2^{\lg^2 n}$	Ω	Ω	Ω
$\lg(n!)$	Ω	Θ	O
$n^{\lg 3}$	Ω	Ω	O

B. Solve the recurrence below using iteration method [2 marks]

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Question 4: Sorting — To be answered in (15) Minutes [ ] / 5 Marks

A. Given the following array {2,5,3,0,2,3,0,3}, the task is to sort them using counting sort

1. *Write the Auxiliary storage (counting array C) after the cumulative sum.*

0	1	2	3	4	5
2	2	4	7	7	8

2. *Pick up the first element to be inserted in the Output array(B) and update B*

1	2	3	4	5	6	7	8
I						3	

3. *Update the Auxiliary storage (counting array C)* **after picking up the first element.**

0	1	2	3	4	5
2	2	4	6	7	8

4. **Update output array (B), after the second element is selected**

1	2	3	4	5	6	7	8
	0					3	

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C. Solve the recurrence below using Master method [2 marks]

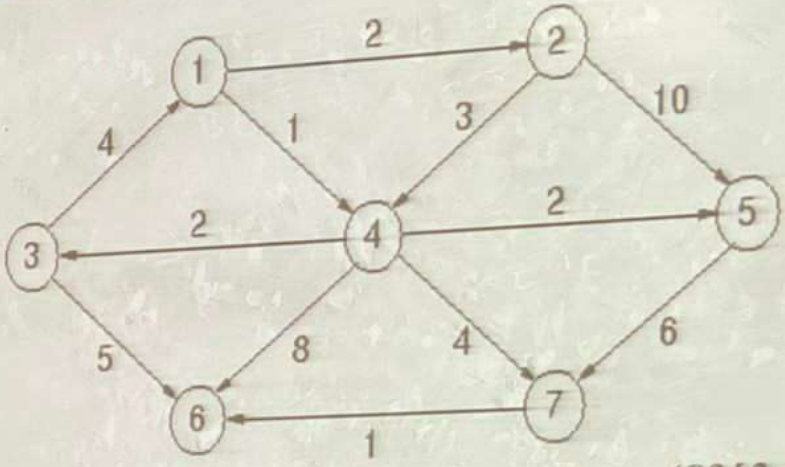
$$T(n) = 2T(n/3) + n^2$$

By Case 3 of the Master Method, we have $O(n^2)$

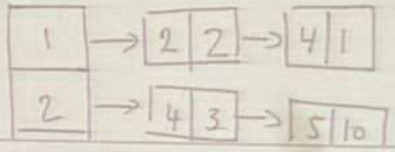
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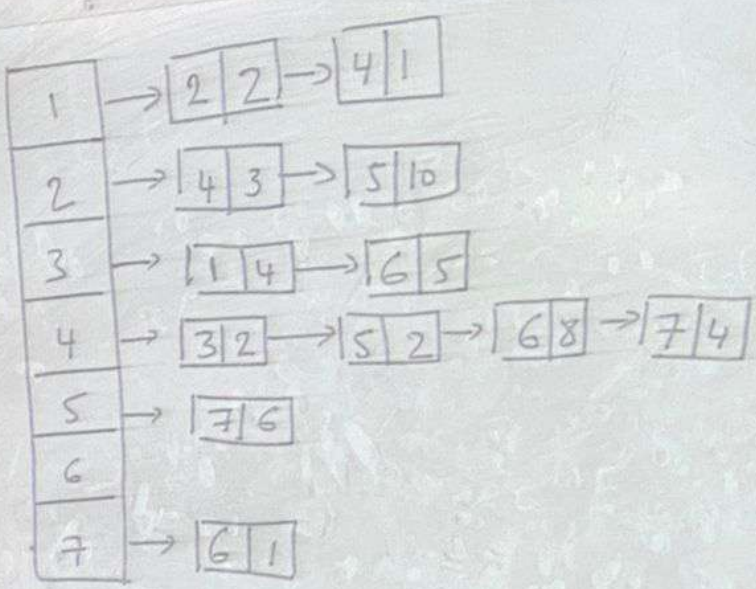
Question 5: *Graph and Dynamic programming* **To be answered in (30) Minutes** [ ] / 13Marks

A. Consider the following directed graph.



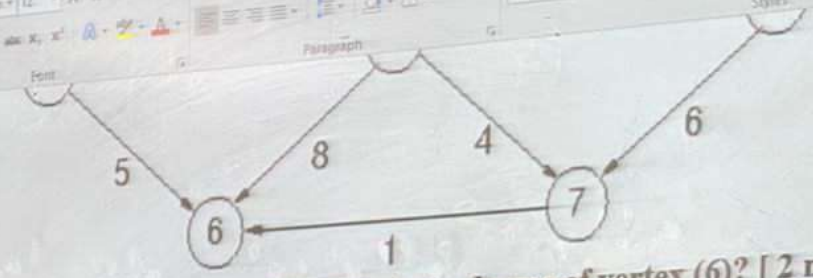
1. **What is the degree of vertex (4)? What is the in-degree of vertex (6)? [2 marks]**
   degree of vertex (4): 6
   in-degree of vertex (6): 3
2. **Show the adjacency-list representation of the above directed graph. [2 marks]**





3. **Apply Dijkstra's algorithm to find the shortest distance between vertex (1) and all other vertices. Fill in the table below. [3 marks]**

Vertex	1	2	3	4	5	6	7
d	0	2	3	1	3	6	5
π	nil	1	4	1	4	7	4

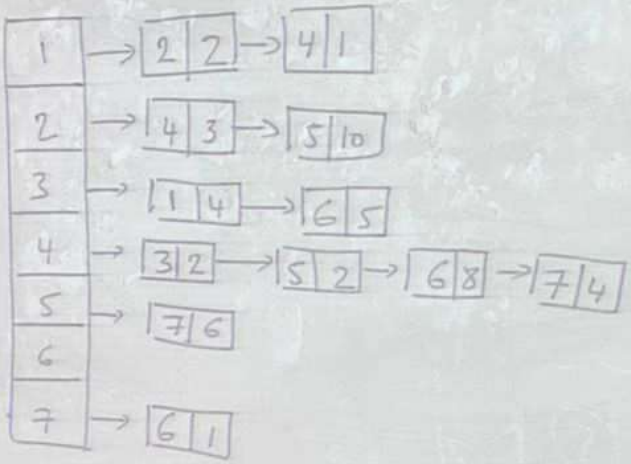


1. **What is the degree of vertex (4)? What is the in-degree of vertex (6)? [2 marks]**

   degree of vertex (4): 6

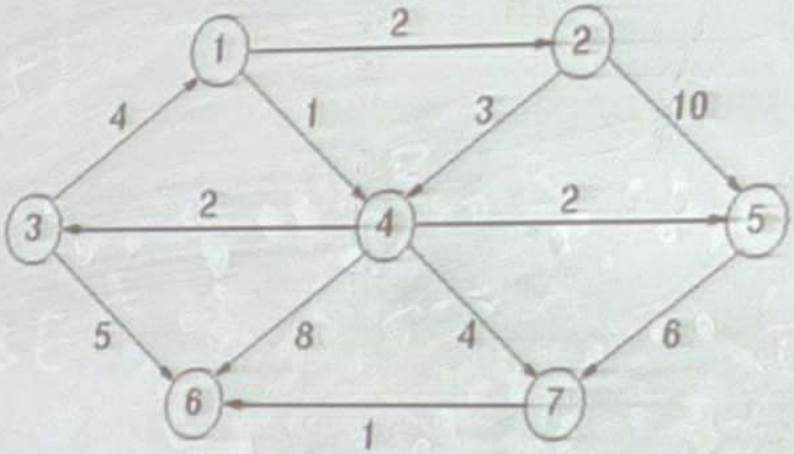
   in-degree of vertex (6): 3

2. **Show the adjacency-list representation of the above directed graph. [2 marks]**

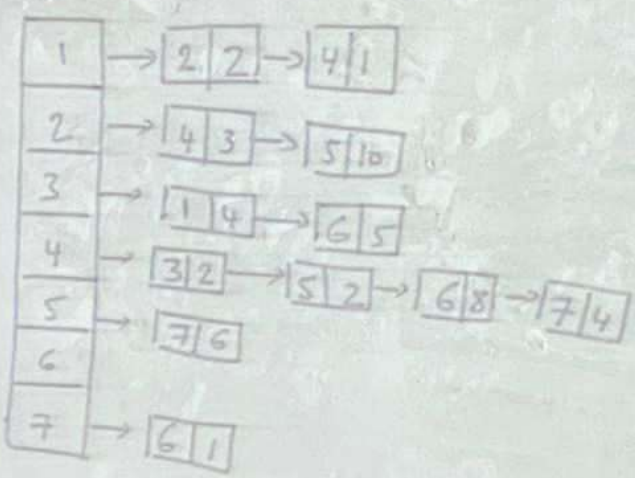


3. **Apply Dijkstra's algorithm to find the shortest distance between vertex (1) and all other**

A. Consider the following directed graph.



1. What is the degree of vertex (4)? What is the in-degree of vertex (6)? [2 marks]
   degree of vertex (4): 6
   in-degree of vertex (6): 3
2. Show the adjacency-list representation of the above directed graph. [2 marks]



3. Apply Dijkstra's algorithm to find the shortest distance between vertex (1) and all other vertices. Fill in the table below. [3 marks]

Vertex	1	2	3	4	5	6	7
d	0	2	3	1	3	6	5
π	nil	1	4	1	4	7	4

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B. A thief has entered a house with a maximum carrying capacity of 5 kg. Inside, there are four items, each with specified weights and values as shown in the table below. What items should the thief choose to maximize his money, considering that he can either take an item entirely or leave it behind entirely? **[4 marks]**

Item	Weight (kg)	Value ($)
Mirror	2	30
Silver nugget	3	40
Painting	4	50
Vase	5	60

Solution:

	0	1	2	3	4	5
0	0	0	0	0	0	0

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[4 marks]

entirely or leave it behind entirely?

Item	Weight (kg)	Value ($)
Mirror	2	30
Silver nugget	3	40
Painting	4	50
Vase	5	60

Solution:

	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	30	30	30	30
2	0	0	30	40	40	70
3	0	0	30	40	50	70
4	0	0	30	40	50	70

The maximum value is: 70 **[1 mark]**

The items that must be put into the knapsack to obtain the maximum value are: Mirror , Silver nugget

[1 mark]

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